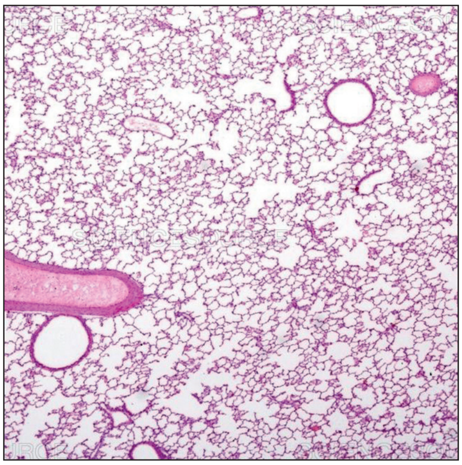
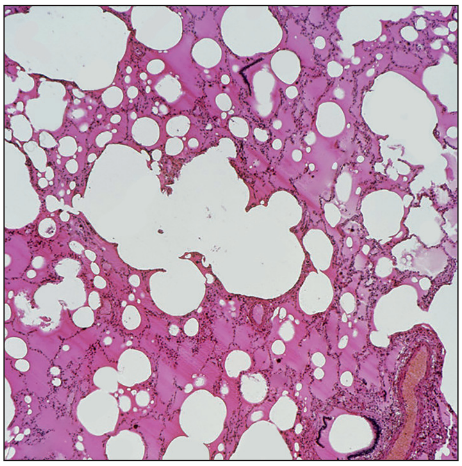
4. Emphysema is a disease of the lungs that results in the breakdown of the walls between adjacent alveoli.

Images 4.1 and **4.2** show sections of lung tissue from an adult with healthy lungs and an adult with emphysema.

Image 4.1 Lung tissue from an adult with healthy lungs	**Image 4.2** Lung tissue from an adult with emphysema
	

(a) (i) Using **Image 4.1** and your own knowledge, describe **and** explain **two** adaptations of the lungs for gas exchange. [2]

(ii) Using **Image 4.2**, explain why the person with emphysema would have a lower oxygen saturation of haemoglobin than an adult with healthy lungs. [2]

Examiner only

(b) The health of the lungs can be assessed by using a device called a spirometer, which measures the rate and depth of inspiration and expiration.

(i) Describe the process of inspiration. [4]

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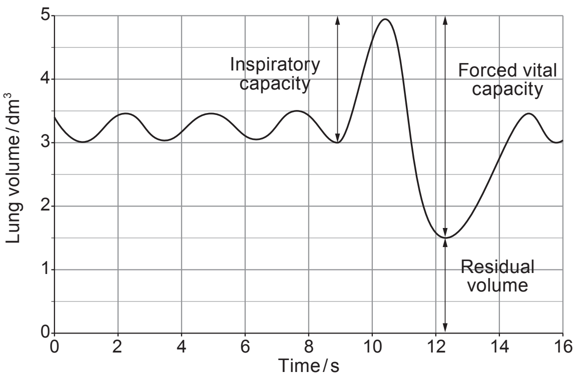
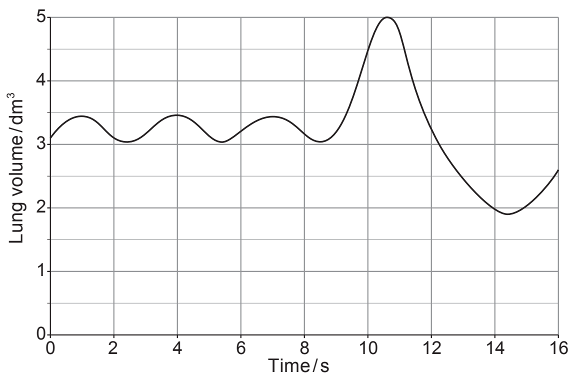
During an investigation an adult with healthy lungs and an adult with emphysema were asked to:

- breathe normally for three breaths
- then to take a very deep breath to fill their lungs, known as a forced inspiration
- then to expire as completely and as rapidly as possible, known as a forced expiration.

Images 4.3 and **4.4** show the spirometry traces produced.

Image 4.3 Spirometry trace from an adult with healthy lungs	**Image 4.4** Spirometry trace from an adult with emphysema
	

Key:

Inspiratory capacity:	Total volume of air that can be taken into the lungs during inspiration.
Forced vital capacity:	Total volume of air that can be forced out of the lungs following a forced inspiration.
Residual volume:	Volume of air that cannot be removed from the lungs.

Emphysema fact file

Emphysema causes the following physical effects on the lungs:

- loss of elastic tissue from the lungs
- thickening of the walls of the bronchioles
- increased mucus production within the bronchioles.

Examiner only

(ii) **Complete Table 4.5** to state **two** differences between the two spirometry traces in **Images 4.3** and **4.4** and use the information from the fact file to explain these differences. [4]

Table 4.5

Difference	Explanation
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....	....

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